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An interpretable short-term electrical load forecasting model based on SHapley Additive exPlanations——A case study in Haidian, Beijing

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ABSTRACT

In recent years, with the rapid development of modern society, the demand for electric power is booming. Accurate forecasting of electrical peak load is crucial for maintaining a stable electricity supply. This study developed daily electrical peak load forecasting models for Haidian by statistical (Stepwise Multiple Linear Regression, SMLR) and machine learning (Extreme Gradient Boosting, XGBoost, and Least Absolute Shrinkage and Selection Operator-Extreme Gradient Boosting, LASSO-XGBoost) algorithms under two scenarios, and conducted an interpretability analysis on the optimal models to quantify the contributions of the critical features for daily electrical peak load through SHapley Additive exPlanations (SHAP) and Partial Dependence Relationship (PDR) analysis. The results showed that XGBoost and LASSO-XGBoost achieved superior predictive accuracy (R^2>0.93), compared with SMLR model. Meteorological factors, past electricity peak load, and holiday parameter were critical features for peak load forecast: Peak load in Haidian was significantly influenced by the "weekend effect", which was notably lower on holidays compared to workdays. Sensible temperature was the most critical meteorological factor for peak load, which increased rapidly when it exceeded 29.5°C. Electricity load exhibited short-term temporal dependence, and previous peak loads were also important contributors to the prediction result. Our results provided valuable insights for the electricity sector in formulating strategies for electricity production and dispatching.

1. Introduction

With the rapid economic growth and population expansion, electricity as a clean and efficient energy source has become increasingly indispensable in modern society [1]. However, unlike other material products, electricity cannot be stored in bulk and must be generated instantaneously to meet demand [2]. Analyzing behavioral patterns and factors influencing electric load, as well as accurately predicting future demand, is critical for effective power generation scheduling and power system management [3]. Overestimating electric load may result in energy wasting and operational costs rising, as reported that if the prediction error is reduced by 1%, a 10 GW power station may save $1.6 million annually [4,5], whereas underestimating load will decrease the reliability of the power system [6]. Therefore, short-term electric load forecasting is a vital process in the planning of the electricity industry and plays a crucial role in the operation of electric power systems.

Short-term electric load prediction is a complex issue due to its dependence on numerous external factors, including weather conditions, human activities, and seasonal variations [7]. At present, there are mainly two types of load forecasting methods in the existing research: One is the traditional forecasting method based on statistical analysis, such as regression and time series analysis. The other is the machine learning based forecasting method, including random forest, long short time memory (LSTM), and extreme gradient boosting model (XGBoost),

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etc [8]. Previous studies developed peak load forecasting models in Beijing based on the multiple linear regression algorithm, indicating that per 1°C increased in the daily minimum air temperature enhanced peak load by 39.7×10^4 kW [9]. However, the statistical models all relied on the assumption of a certain probabilistic statistical relationship among variables, which exhibit limitations in handling dynamic non-deterministic relationships (e.g., stochastic fluctuations or time-varying parameters). In recent years, traditional machine learning and deep learning are increasingly being used to energy forecasting. For example, Imani [10] used Convolutional Neural Network (CNN) to extract the nonlinear relationships of the residential load and achieved highly accurate results. Li et al. [11] developed a peak load forecasting model for Beijing using BP neural network algorithm, achieved a correlation coefficient exceeding 0.95 between predicted and observed values. Considering that the temporal dependence of the actual load is reflected in dynamics and variability, an improved series-to-sequence gated recurrent unit network (S2S-IGRU) model is proposed by Li et al. [12], which enhanced the robustness of short-term load forecast. Empirical studies suggested that coupling decomposition methods (e.g., EMD, CEEMDAN) with predictive models can improve forecasting precision. Fan et al. [13] combined Empirical Mode Decomposition (EMD) with support vector regression (SVR) to predict the electricity load, and the result showed that the proposed EMD-SVR outperformed the baseline models on electricity load forecasting. Ran et al. [14] proposed a hybrid model incorporating Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), sample entropy (SE) and Transformer (TR) model (CEEMDAN-SE-TR), which get the best forecasting results of all the comparison models. In recent years, with the rapid advancement of artificial intelligence, non-linear intelligent methods have been increasingly applied to electric load forecasting. For example, Li et al. [11] developed an electrical peak load forecasting model for Beijing using a BP neural network algorithm, achieved a correlation coefficient exceeding 0.95 between predicted and observed values, while the influence pattern of critical features of peak load was still unknown. As reported, machine learning has great potential for improving products, processes and research, while its lack of interpretability is a barrier to it adoption [15]. To address this limitation, SHapley Additive exPlanations (SHAP) theory [16] and partial dependency analysis [17] have been introduced for interpretable machine learning (IML) and have been widely proven to be effective [18]. Due to rapid change in electrical consumption patterns, it is important for forecasting models to learn and interpret the load consumption from historical data concerning the data changes with respect to the associated factors [19].

Thus, in this study, we conducted a case study in Haidian District, constructed daily electrical peak load forecasting models based on traditional statistical algorithm and machine learning algorithms to achieve better performance. Furthermore, we employed SHAP and PDR analysis to provide interpretable analyses on the optimal models, quantifying the influence of critical features on the prediction results. Our findings offer valuable insights for electric power generation and dispatching in the region, contributing to more efficient and reliable power system management. This publication makes the following contributions:

1). This study developed advanced forecasting frameworks by XGBoost and LASSO-XGBoost algorithms, incorporating probability density estimation to enhance the precision of daily peak load predictions. This methodological innovation significantly improves forecasting accuracy compared to conventional approaches.
2). Our work introduced a comprehensive interpretability analysis framework combining SHAP and PDR analysis, quantitatively evaluated feature contributions: including meteorological parameters, historical load, and holiday effects.
3). Our findings offer substantial decision-making support for power grid operations for optimizing electricity generation scheduling and dispatching strategies, by introducing accuracy peak load prediction and the tipping points of critical features.

The remainder of this paper is arranged as follows: Section 2 addresses materials and methods for model construction and interpretability analysis. Section 3 presents the model construction plan. Section 4 and 5 elaborate on the results and discussions, respectively. Section 6 concludes the study with key findings and future work.

2. Materials and methods

2.1. Study area

Haidian, located in the northwest of Beijing, China, is a hub of technological innovation, higher education, and high-tech industries. The economic and social development of Haidian largely relies on the electrical supply. Taking 2022 as an example, Haidian's GDP exceeded 1 trillion yuan, which is the first district-level administrative region in China surpassing this mark, with consumed 16.72 billion kWh electricity [20]. Thus, we set a case study in Haidian, conducting interpretability short-term electrical load forecasting models to achieve the accurate load forecasting and quantify the impacts of important factors.

2.2. Data

The daily electrical peak load data of Haidian District from 2022 to 2023 was provided by the State Grid Beijing Haidian Electric Power Supply Company. And daily meteorological data during the same period was provided by Haidian Meteorological Bureau, including minimum temperature, maximum temperature, average humidity, minimum humidity, sunshine hours, average wind speed, extreme wind speed, average air pressure, cumulative precipitation.

Besides, the sensible temperature (ST) was also introduced as an influence factor, which was calculated by:

$$ST = 1.07T + 0.2e - 0.65V - 2.7 \tag{1}$$

$$e = \frac{RH}{100} \times 6.105 \times e^{\frac{17.27T}{237.7+T}} \tag{2}$$

Where T is air temperature (°C), e is vapor pressure (hPa), V is wind speed(m/s), RH is relative humidity (%).ST was introduced in models as a metrological factor.

2.3. Extreme Gradient Boosting

Extreme Gradient Boosting (XGBoost) is a scalable machine learning system for tree boosting, which core idea is to build a powerful prediction model by combining multiple decision trees [21]. It can reduce the overfitting and achieve high accuracy by introducing a regularization term in the loss function and using a second-order Taylor expansion to approximately fitting the loss function [22]. XGBoost has proven highly effective for power load simulation research [23]. XGBoost uses K additive functions to predict the output:

$$\hat{y}_i = \phi(x_i) = \sum_{k=1}^{K} f_k(x_i),\ f_k \in \mathscr{F} \tag{3}$$

Where $\mathscr{F}$ is the space of regression trees, q represents the structure of each tree that maps an example to the corresponding leaf index, T is the number of leaves in the tree. Each f_k corresponds to an independent tree structure q and leaf weights ω.

The objective function of the XGboost algorithm is.

$$\mathscr{L}(\phi) = \sum_{i=1}^{N} l(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^{K} \Omega(f_x) \tag{4}$$

Where l is a differentiable convex loss function that measures the difference between the prediction $\hat{y}_i$ and the target y_i. Ω penalizes the complexity of the model and is calculated as $\Omega(f) = \gamma T + \frac{1}{2}\lambda \| \omega \|$, T denotes the number of leaf nodes, ω denotes the leaf node weights, λ denotes the regularization penalty term coefficient, and γ is the penalty factor for the number of leaf nodes.

The minimization of the objective function is achieved through continuous iterations, and the objective function after each iteration is:

$$\mathscr{L}^{(t)} = \sum_{i=1}^{N} l\left[y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right] + \Omega(f_t \tag{5}$$

The objective function is expanded as follows:

$$\mathscr{L}^{(t)} \approx \sum_{i=1}^{N} l\left[y_i, \hat{y}_i^{(t-1)} + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)\right] + \Omega(f_t) \tag{6}$$

Where $g_i = \partial_{\hat{y}_i^{(t-1)}} l\left(y_i, \hat{y}_i^{(t-1)}\right)$ and $h_i = \partial^2_{\hat{y}_i^{(t-1)}} l\left(y_i, \hat{y}_i^{(t-1)}\right)^2$ are first and second order gradient statistics on the loss function.

In order to get the minimum value of the objective function and make its derivative equal to 0, the optimal weight ω_j^* of leaf j is:

$$\omega_j^* = -\frac{\sum_{i \in I_j} g_j}{\sum_{i \in I_j} h_j + \lambda} \tag{7}$$

and the corresponding optimal value is calculated by

$$\widetilde{\mathscr{L}}^{(t)}(q) = -\frac{1}{2} \sum_{j=1}^{T} \frac{\left(\sum_{i \in I_j} g_j\right)^2}{\sum_{i \in I_j} h_j + \lambda} + \gamma T \tag{8}$$

In this study, the XGBoost algorithm is implemented using the "xgboost" package (version 1.7.8.1) in R-4.3.3.

2.4. Least Absolute Shrinkage and Selection Operator

The Least Absolute Shrinkage and Selection Operator (LASSO) is implemented to simultaneously identify salient features and induce sparsity-based dimensionality reduction in the dataset [24]. The principle of LASSO is to impose a penalty on the model's coefficient on the basis of the ordinary least-squares estimation.

In this study, the LASSO-XGBoost model was built by selecting feature subset based on LASSO, and set the subset as the input to the XGBoost model.

2.5. Stepwise Multiple Linear Regression

Stepwise Multiple Linear Regression (SMLR) models the relationship between the dependent variable and explanatory variables through a system of linear equations. In this study, the analysis employed bidirectional stepwise selection with Akaike's Information Criterion (AIC), initiating from a full linear model containing all predictors [25]. Through iterative forward selection and backward elimination, variables were dynamically adjusted based on their AIC impacts: incorporating those yielding maximum AIC reduction while removing contributors causing minimal AIC elevation. The process terminated when no single-variable modification improved model fit, resulting in an optimized predictor set balancing parsimony and predictive capacity. Hierarchical hypothesis testing (α=0.15) controlled false discovery rates during model stabilization. The AIC can be calculated as:

$$\mathrm{AIC} = \left(\log \frac{RSS}{n}\right) + 2kn \tag{9}$$

where RSS is the sum of squared errors, k is the number of variables in the model, and n is the number of samples.

2.6. Kernel density estimation

Kernel Density Estimation (KDE) is a non-parametric density estimation method that does not require prior assumptions about the distribution form. It derives more accurate probability density estimation results by relying solely on the inherent distribution of the sample data [26]. The KDE is estimated as:

$$\widehat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{10}$$

where $K(\cdot)$ is the kernel function, h is the bandwidth of the kernel function, and n is the sample size. In this study, KDE was used to calculate the confidence intervals of predicted values.

2.7. Validation/accuracy assessment

The Root Mean Square Error (RMSE) reflects how far off the predicted values on average are from the observed values, Mean Relative Error (MRE) is calculated as the mean of absolute percentage errors between observed and predicted values, and R-squared (R^2) measures the closeness of the predicted and actual values to the 1:1 line, which are calculated as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{\mathrm{n}} \sum_{\mathrm{i}=1}^{\mathrm{n}} \left(y_i - \hat{y}_i\right)^2} \tag{11}$$

$$\mathrm{MRE} = \frac{1}{\mathrm{n}} \sum_{\mathrm{i}=1}^{\mathrm{n}} \frac{y_i - \hat{y}_i}{y_i} \times 100\% \tag{12}$$

$$R^2 = 1 - \frac{\sum_{i=1} \left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n} \left(y_i - \overline{y_i}\right)^2} \tag{13}$$

where n is the number of observations; $\hat{y}_i$ and y_i are the predicted and actual values of electrical peak load.

The Continuous Ranked Probability Score (CRPS) is a metric for evaluating probabilistic forecasts by quantifying the discrepancy between the predicted cumulative distribution function (CDF) and the observed CDF:

$$CRPS = \sum_{\mathrm{i}=1}^{\mathrm{n}} \left(F_t(\tau_i) - \varepsilon\left[\tau_i \geq y_t^{(i)}\right]\right)^2 \tag{14}$$

where $F(\tau_i)$ is predicted CDF, y_t is observed values, ε is indicator function.

2.8. SHapley Additive exPlanations and Partial Dependence Relationship

SHapley Additive exPlanation (SHAP) is a theoretical approach used to explain the outcomes of machine learning models [16]. The SHAP value for each feature measures its contribution to the model's output, which is calculated by:

$$\varnothing_i(\mathrm{x}) = \sum_{S \subseteq N\{i\}} \frac{|S|!(8 - |S| - 1)!}{8!} \left[f_x(S \bigcup N\{i\}) - f_x(S)\right] \tag{15}$$

where $\varnothing_i(\mathrm{x})$ represents the SHAP values of the feature for sample x, N is the set of all input features, S is the set of non-zero indices in x, and f is the prediction model to be explained. Based on the SHAP values of all the given samples of each feature, the average of absolute values of SHAP is calculated to draw the importance map to explain the importance of that feature in the prediction models. And the effect of each important feature is explained by plotting the SHAP dependency plot (SDP).

Partial Dependence Relationship (PDR) demonstrates the marginal impact of important features on the model's predicted outcomes, which

is defined as:

$$\widehat{f}_{x_s}(x_s) = E_{x_C}[\widehat{f}(x_s, x_C)] = \int \widehat{f}(x_s, x_C) d\mathbb{p}(x_C) \quad (16)$$

Where $\widehat{f}_{x_s}(x_s)$ is the mean dependent variable computed by x_s, x_s is the target independent variable, x_C is the complementary sets of x_s, $\mathbb{p}(x_C)$ is the joint probability density of all independent variables.

In this study, the independent variable resolution is set as 0.1 (the minimum step size of meteorological observations) to plot the partial dependence plot (PDP) to systematically evaluate covariate-effect relationships of Forecasting Models with SDP.

3. Modeling framework

3.1. Predictive features

As reported, meteorological factors significantly influenced electrical load, with impacts stemming not only from current weather conditions but also from the prolonged effects of prior's [27]. Additionally, daily load profiled exhibit substantial variations driven by people's daily routines [28]. Therefore, historical electrical load and holiday parameter served as critical indicators, reflecting the inertia of user consumption behavior [8]. In this study, historical and current meteorological factors, past electricity peak load, and holiday parameter were selected as predictive features in forecast models.

3.2. Modeling plan

In this study, daily electrical peak load models were built by SMLR, XGBoost and LASSO-XGBoost algorithm and two modeling scenarios were set: Scenario 1, introducing meteorological factors and holiday parameter; Scenario 2, introducing meteorological factors, past electricity peak load, and holiday parameter (Table 1). For each model, 80% of the data was randomly selected as the training set for modeling and the remaining 20% was used as the test set for model verification, and repeated for three times for each scenario. To verify the applicability of the algorithms, this study simulated power load based on the December 2011 data of the GEFCom2014 - L_V2 dataset [29], which achieved favorable results (details see Supplementary). The flow chart of model development and interpretability analysis is drawn in Fig. 1 and nomenclature is list in Table 2.

Table 1
Modeling scenarios.

Element	Content	Scenario 1	Scenario 2
Meteorological factors	**The intraday, one day, two days, and three days before:**	√	√
	Daily average, minimum, maximum air temperature		
	Daily average, minimum, maximum surface temperature		
	Daily average, minimum humidity		
	Daily sunshine hours		
	Daily average, extreme wind speed		
	Daily average air pressure		
	Daily accumulated precipitation		
	Daily average sensible temperature		
Date property	Workday or holiday	√	√
Previous electrical loads	**One day, two days, and three days before:**		√
	Daily electrical peak load		

4. Results

4.1. Characteristics of daily electrical peak load in Haidian

Daily electrical peak load ranged from 1609.97 MW to 3687.30 MW, and the highest peak occurred in July and August, reaching 3500 MW, followed by a secondary peak in December and January, with exceeding 2300 MW. The lowest loads were observed in April, May, and September, with values below 2000 MW. Daily electrical peak load during workdays varied in 1763.21～3687.30 MW and averaged at 2510.92 MW, while that during holidays ranged from 1609.97 MW to 3250.41 MW and averaged at 2249.33 MW. The electrical peak load during holidays was lower than that on workdays during the same period, with an average reduction of 11.62%. (Fig. 2).

The box plot shows the maximum, minimum, median, and quartiles of electrical peak load.

4.2. Performance of different models for simulating daily electrical load

In Scenario 1, XGBoost and LASSO-XGBoost (average R^2=0.92, RMSE≈120 MW, MRE≈3.9%, CRPS≈430 MW) performed good in simulating peak load. In Scenario 2, the accuracy of XGBoost and LASSO-XGBoost (R^2=0.95, RMSE≈93 MW, MRE≈2.9%, CRPS≈240 MW) both improved (Table 3). According the optimal models of Scenario 1 and 2, almost all predicted values dropped within the 95% coefficient band (Fig. 3). In contrast, the SMLR models exhibited relatively large bias in Scenario 1, with averaged R^2, RMSE, MRE and CRPS were 0.68, 243.41 MW, 7.92% and 1326.02 MW. And their performance improved markedly in Scenario 2, with averaged R^2, RMSE, MRE and CRPS were 0.92, 124.82 MW, 3.96% and 371.28 MW (Table 3).

4.3. Interpretability analysis of the critical features for daily electrical peak load

In Scenario 1, the intraday sensible temperature was the most critical feature for daily electrical peak load, with the averaged absolute values of SHAP values (Mean(|SHAP|)) of 128.53, followed by the date property indicating whether it was a holiday or not, with the Mean(|SHAP|) of 89.59. The intraday surface temperatures also played notable roles, with Mean(|SHAP|) ranged from 74.85 to 26.06 (Fig. 4 a). In Scenario 2, peak load of one day before contributed the most, with the Mean(|SHAP|) of 195.37, and the holiday parameter ranked the second, and its Mean (|SHAP|) was 78.39. Notably, the intraday sensible temperature remained the most important meteorological factor, with the Mean (|SHAP|) of 49.92 (Fig. 4 b).

4.4. Impacts of the critical features on daily electrical peak load

In Scenario 1, when the intraday sensible temperature was below 24.8°C, it exhibited a weak negative effect. When it ranged from 24.9°C to 28.2°C, it showed a weak positive effect, with an averaged SHAP value of 18.9 and a partial dependency value ranged from 15.5 MW to 17 MW. When the sensible temperature exceeded 28.2°C, its positive contribution to the peak load increased sharply, with the SHAP value raised from 70 to 500 with the sensible temperature climbed from 28.2°C to 36°C (Fig. 5 a). Holidays demonstrated a significantly negative effect on peak load (Fig. 5 b). The influence of average surface temperature on peak load follows a distinct pattern, its positive contribution escalated rapidly when it dropped below 9.7°C, and peaked within the range of -2.6°C to 3.6°C (Fig. 5 c). The intraday minimum surface temperature exhibited a strong positive contribution to peak load when exceeded 14.7°C, but a weak negative contribution when it fell below 14.7°C (Fig. 5 d).

In Scenario 2, previous peak loads were the most significant contributors to the intraday electrical peak load, whose impacts changed from negative to positive and intensify with them increases. The peak

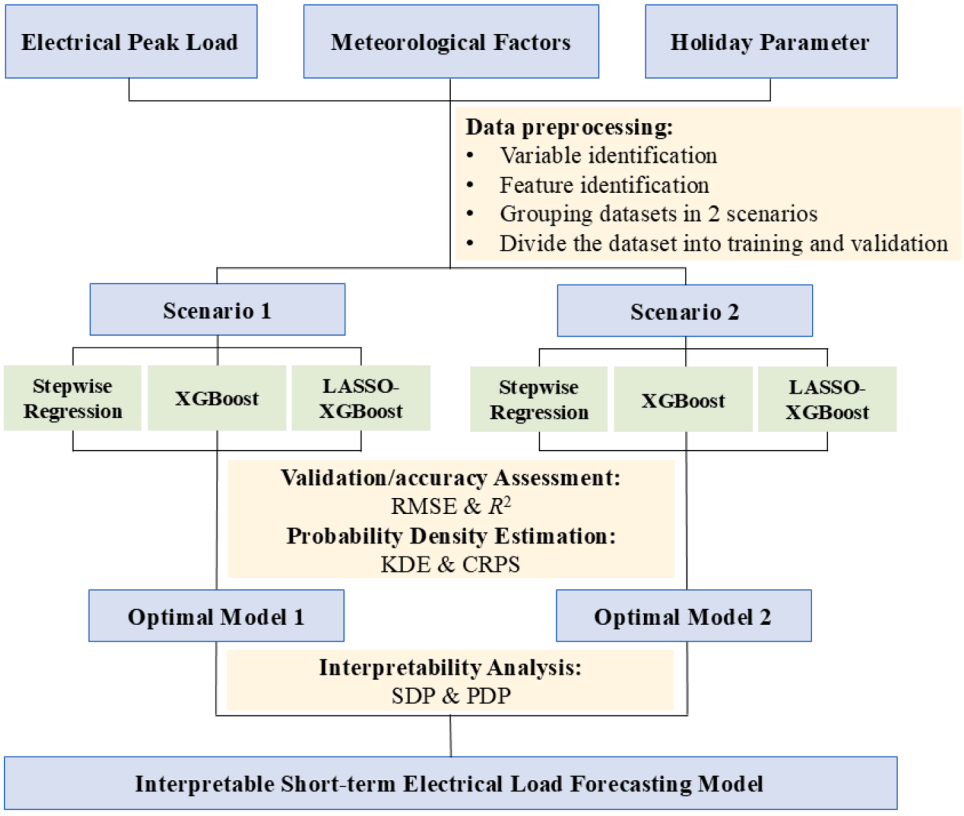


Fig. 1. Schematic diagram of model construction and interpretability analysis.

Table 2
Nomenclature.

Abbreviation	Definition
XGBoost	Extreme Gradient Boosting
LASSO	Least Absolute Shrinkage and Selection Operator
SMLR	Stepwise Multiple Linear Regression
KDE	Kernel Density Estimation
RMSE	Root Mean Square Error
R^2	R-squared
CRPS	Continuous Ranked Probability Score
SHAP (SDP)	Shapley Additive explanation (SHAP Dependency Plot)
PDR (PDP)	Partial Dependence Relationship (Partial Dependence Plot)

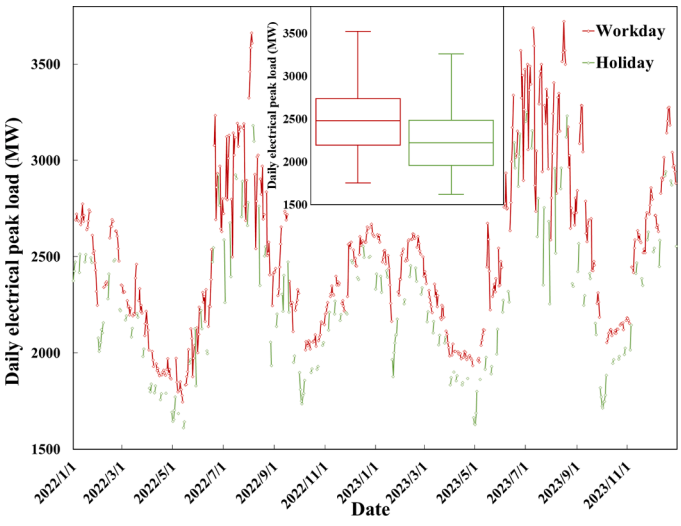


Fig. 2. Intra-annual characteristics of daily electrical peak load in Haidian.

load of one day before showed a negative effect when it was below 2329.2 MW, a weak positive effect within the range of 2330.1~2385.1 MW, with an averaged SHAP value of 8.07, and a strong positive effect when it exceeded 2385.1 MW (Fig. 6 a, according to the results of SDP). The peak load of three days before exhibited a similar influence pattern, though its contribution was only about 25% of that of one day before (Fig. 6 d). Holidays demonstrated negative contributions to the electrical peak load compared to workdays (Fig. 6 b). The intraday sensible temperature remained the most important meteorological factor, and it displayed a weak positive effect below -10.4°C (with averaged SHAP and partial dependency values of 0.67 and 5.70 MW), a weak negative effect within the range of -10.4~26.1°C (with averaged SHAP and partial dependency values of -35.85 and -33.30 MW), a weak positive effect between 26.2~29.5°C (with averaged SHAP and partial dependency values of 10.36 and 16.28 MW), and a strong positive contribution when exceeding 29.5°C (Fig. 6 c).

5. Discussion

5.1. Models evaluation

In Scenario 1, XGBoost and LASSO-XGBoost performed better than SMLR model. This is fundamentally caused by the models' differential capacity to capture dynamic nonlinear relationships, as the tree-based algorithms inherently accommodate complex meteorological-load interactions, while the linear parametric framework is inadequate for modeling such nonlinear dynamics [30].

In Scenario 2, all models produced measurable improvements, especially for the SMLR models. As electrical peak load typically exhibited inertial characteristics, which had a strong self-correlation. The correlation coefficient between the daily peak load and its value of one day before was 0.921. Despite this enhancement, the SMLR predictions remained statistically weaker than ensemble methods. This

Table 3
Prediction accuracy of daily electrical peak load.

Model	Test	Scenario 1				Scenario 2			
		R^2	RMSE (MW)	MRE (%)	CRPS (MW)	R^2	RMSE (MW)	MRE (%)	CRPS (MW)
XGBoost	I	0.92	125.18	4.19	566.99	0.97	81.00	2.68	266.67
	II	0.91	124.89	4.06	387.00	0.96	93.47	2.87	151.42
	III	**0.94**	**111.56**	**3.62**	**334.50**	0.94	106.59	3.19	246.07
	Average	0.92	120.54	3.96	429.50	0.95	93.69	2.91	233.22
LASSO-XGBoost	I	0.92	122.87	4.07	543.79	**0.97**	**79.87**	**2.63**	**160.65**
	II	0.91	124.41	4.00	418.86	0.96	89.36	2.77	186.92
	III	0.94	114.14	3.64	339.37	0.93	109.47	3.33	314.82
	Average	0.92	120.48	3.90	434.01	0.95	92.90	2.91	242.30
SMLR	I	0.68	245.90	8.07	1477.35	0.92	123.10	3.82	254.69
	II	0.68	235.55	7.75	1270.66	0.92	125.19	3.99	354.48
	III	0.68	248.78	7.95	1230.05	0.91	126.16	4.06	504.67
	Average	0.68	243.41	7.92	1326.02	0.92	124.82	3.96	371.28

Note: the optimal modeling results were bolded.

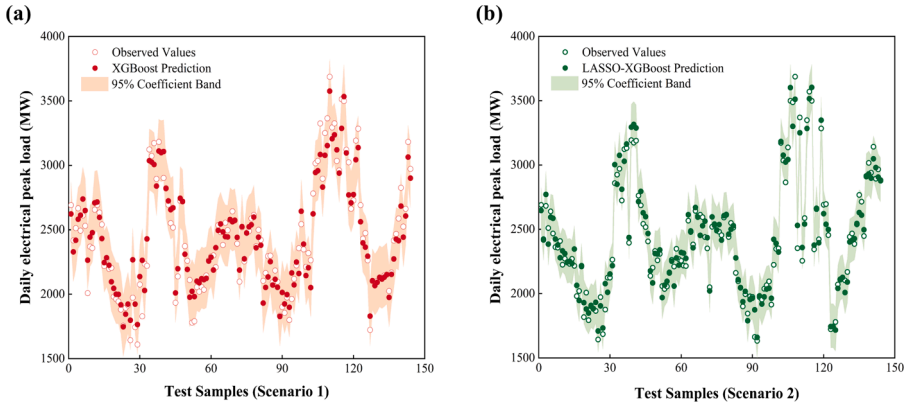


Fig. 3. Results of optimal model based on Scenario 1 (a) and Scenario 2 (b).

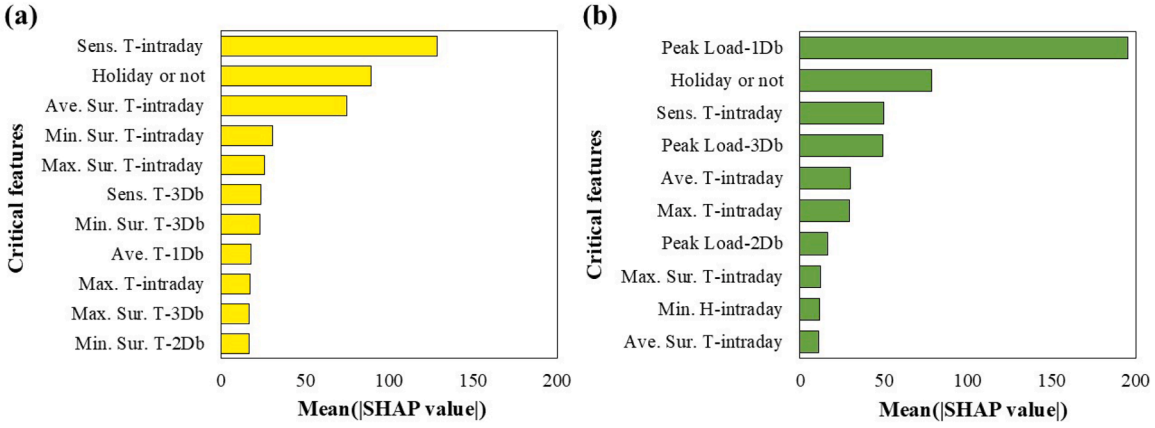


Fig. 4. Importance map of SHAP features of optimal models of Scenario 1 (a) and Scenario 2 (b). (Sens.: sensible, T: temperature, Sur.: surface, H: humidity, Ave: average, Max.: Maximum, Min: minimum, 1Db: 1 day before, 2Db: 2 days before, 3Db: 3 days before).

pattern revealed two key findings: First, temporal autocorrelation constituted an essential predictive feature for peak load forecasting, as evidenced by the improvements in Scenario 2 compared to Scenario 1 [31]. Second, Model architecture dictated accuracy limits, with nonlinear methods inherently superior in capturing complex dynamics [32].

5.2. Impacts of critical features

In both two Scenarios, sensible temperature was the most important meteorological factor, which is considered as better indicative than any other single meteorological factor in electrical load forecasting [33]. The peak load increased sharply with the increase of sensible temperature when it exceeded 28.2°C in Scenario 1, and 29.5°C in Scenario 2. The different threshold could be primarily attributed to the inconsistency of data sources. In Scenario 2, the incorporation of historical electrical load enhanced the data features, thereby improved the model's predictive accuracy and the tipping point of 29.5°C was considered as more credible. This finding was consistent with previous study [34].

Low temperature may also enhance electricity demand [35]. As shown in Fig. 5 c, when the surface temperature dropped to 3.6°C, its positive contribution to peak load increased rapidly with its decrease;

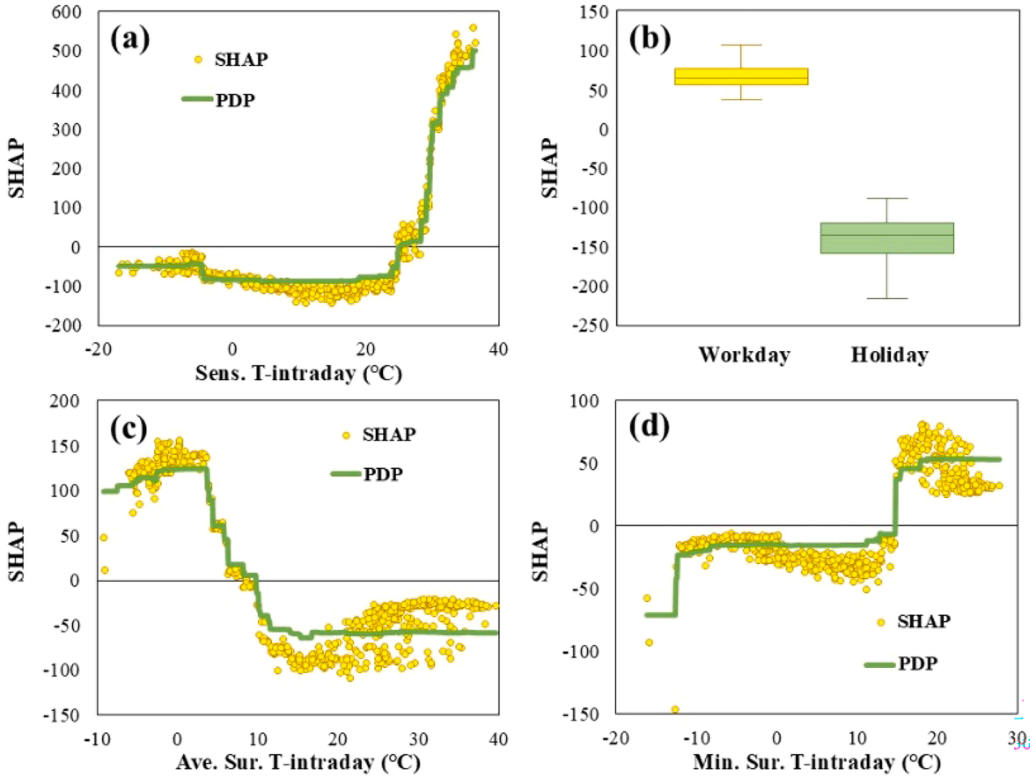


Fig. 5. SHAP Dependency Plot (SDP) and Partial Dependency Plots (PDP) of the optimal model of Scenario 1 (Top 4 important features). Intraday sensible temperature (a), holiday or not (b), intraday average surface temperature (c) and intraday minimum surface temperature (d). The box plot (b) shows the maximum, minimum, median, and quartiles of electrical peak load.

while with it further cooling down to -2.6°C and below, its positive contribution decreased slightly. This phenomenon is largely caused by the Beijing's centralized heating, which turns on at daily average air temperature below 5°C. As surface temperature in winter was lower than air temperature, as a result, the critical value of surface temperature (-2.6°C) is lower than 5°C. Thus, the peak demand on electric power heating was occurred at relatively low temperature.

In addition, even though the samples on workdays and holidays were not modeled respectively, we still found that holiday exhibited a significant negative impact on peak load. Our results were in good agreements with previous studies [29]. It proved that XGBoost and LASSO-XGBoost can satisfy the need for sample classification by providing accurate labeling information.

5.3. SHAP and PDR

Interpretability analysis was conducted by SHAP and PDR. SHAP values were calculated by actual independent variables, which were discrete and we could hardly get a smooth curve to obtain the accurate thresholds of independent variables. Whereas the PDR has the advantage to obtain relatively smooth and continuous curves based on the defined input [36]. In this study, our comparative analysis revealed notably divergent outcomes between SHAP and PDR when quantifying the historical electrical load contributions to the predictions (Fig. 6 a, d). As the PDR might produce substantial estimation biases when handling strongly correlated independent variables [15]. In Scenario 2, the multicollinearity between historical electrical loads and meteorological variables fundamentally compromised the PDR's assumption, thereby resulting in significant deviations. The superior performance of SHAP in this context could be attributed to its game-theoretic foundation, which systematically accounts for feature interactions through permutation-based coalition evaluation. SHAP enables to accurately captures nonlinear relationships and provides locally faithful explanations [37,38].

5.4. Limitations and future work

In this study, we focused on the short-term electrical load forecasting based on the daily data from 2022 to 2023, and achieved good predictions. While, on a perspective of long-term forecasting, long-term data can reveal slow-varying relationships between load and macro-economic factors (e.g., Gross Domestic Product (GDP), population growth, education level), policies (e.g., carbon reduction targets), or technological advancements (e.g., Electric Vehicle adoption), also incorporates enhanced information on periodicity (e.g., seasonal and yearly cycles) and abrupt changes (e.g., extreme weather). Besides, electricity price is also critical feature for peak load prediction. This study was under the assumption that the aforementioned factors remained constant. However, with accessing to longer time-series data and by comprehensively studying the load change patterns in different seasons and years, as well as their associations with macro-economic factors and policy orientations, will further enhance the prediction result and provide strong support for the long-term planning and resource allocation of power systems [5,39].

This study showed that the XGBoost model can effectively predict short-term loads using only meteorological factors. Subsequent research can delve deeper into the more in-depth integration of it with trend-based algorithms such as LSTM. Referring to the analysis of multiple models in Akhtar et al. [40], different models have their respective advantages and disadvantages. It is expected that through combination, the prediction accuracy and stability can be enhanced.

The importance rankings and impact thresholds of key factors was identified in this study, which are of great significance for dynamic load allocation strategies. Further research can be conducted on how to more precisely utilize the rankings and thresholds of critical factors to dynamically adjust load allocation under fluctuations of environmental

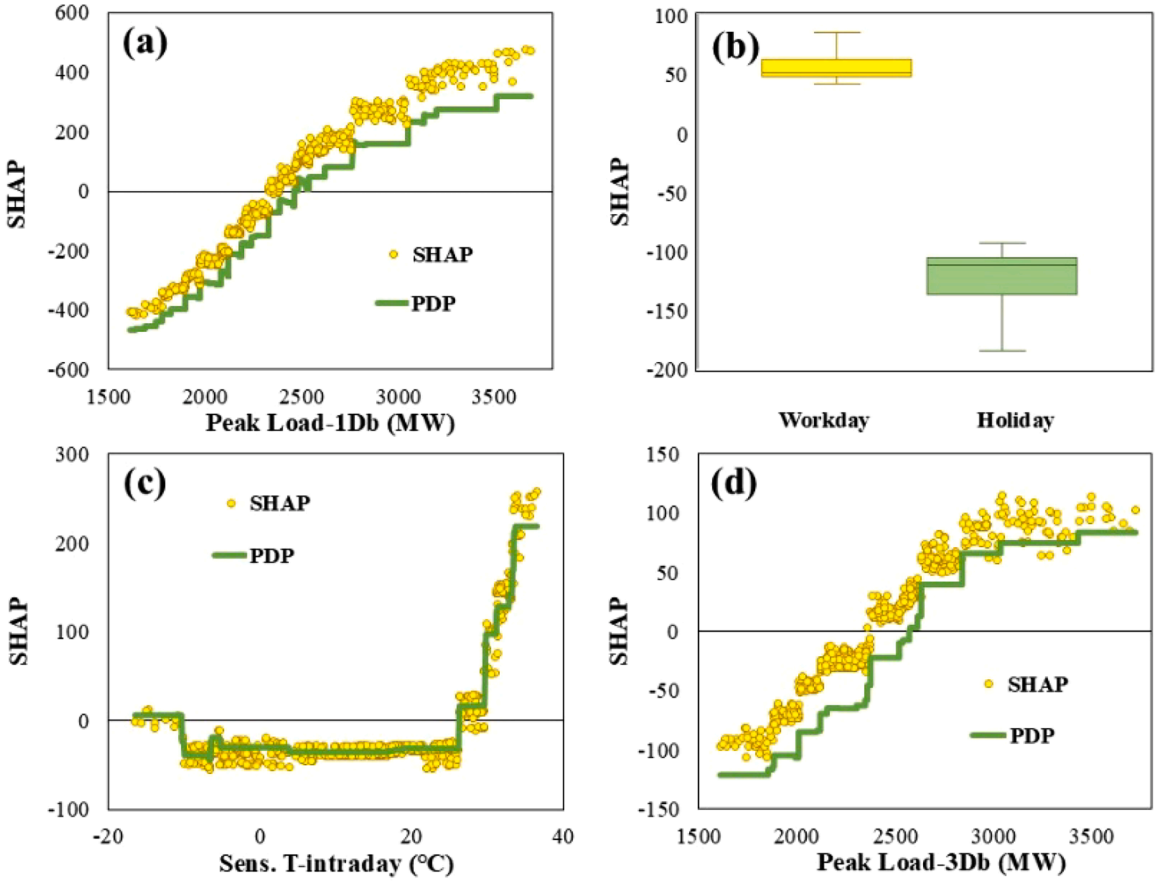


Fig. 6. SHAP Dependency Plot (SDP) and Partial Dependency Plots (PDP) of the optimal model of Scenario 2 (Top 4 important features). Peak load of 1 day before (a), holiday or not (b), average surface temperature of the day (c) and peak load of 3 days before (d). The box plot (b) shows the maximum, minimum, median, and quartiles of electrical peak load.

parameters [41]. For example, when there is a sudden change in temperature or during special holidays, according to the importance of critical factor, power supply to key areas or users should be prioritized, and at the same time, the overall power allocation should be optimized to improve the operational efficiency and reliability of the power system.

6. Conclusion

In this study, the interpretability electrical peak load models for Haidian were built by different algorithm under two scenarios. XGBoost and LASSO-XGBoost performed better in simulating the daily electrical peak load than SMLR algorithm. Holidays demonstrated significant negative effects on electrical peak load. Sensible temperature was identified as the most critical meteorological factor influencing electrical peak load. Electricity load exhibited short-term temporal dependence, and previous electrical loads were important contributors to the intraday electrical peak load. However, the study was limited by the relatively short timeframe of the research data (2021–2022), which prevented the models from accounting for the impacts of social factors such as urban development, economic growth, and population increase. Future research should focus on long-term electrical load simulations with enriched datasets.

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CRediT authorship contribution statement

Jialin Wang: Writing – review & editing, Writing – original draft, Conceptualization. **Bo Yu:** Supervision. **Xiao Chen:** Methodology. **Guisheng Dai:** Data curation. **Gaoju Dai:** Formal analysis. **Wenjun Liu:** Formal analysis. **Na He:** Supervision. **Pingping Zhu:** Data curation. **Zhaoyin Yin:** Methodology, Conceptualization. **Zhihua Pan:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.epsr.2025.111769.

Data availability

The data that has been used is confidential.

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